

Day_1__EDA Fundamentals

September 7, 2026

1 Day 1 - EDA Fundamentals

1.1 Objective

The Objective of this task is to perform basic Exploratory Data Analysis (EDA) On the Customer Churn Dataset.

In this task, we will examine: - Dataset Shape - Data types - Missing Values - Descriptive statistics

```
[2]: import pandas as pd

df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)
df
```

```
[2]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	
14	1015	30	15	58	Month-to-month	
15	1016	47	40	90	Two year	
16	1017	22	4	45	Month-to-month	
17	1018	39	32	80	One year	
18	1019	53	54	100	Two year	
19	1020	33	16	70	Month-to-month	

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No
3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

1.2 1. Dataset Shape

The shape of the dataset shows the total number of rows and columns present in the dataset.

```
[3]: df.shape
```

```
[3]: (20, 9)
```

1.3 2. Data Types

Cheacking the data type of each column hepls us identify numerical and categorical variables.

```
[4]: df.dtypes
```

```
[4]: CustomerID      int64
Age                int64
Tenure_Months      int64
MonthlyCharges     int64
Contract           str
InternetService    str
TechSupport        str
PaymentMethod      str
Churn              str
dtype: object
```

1.4 3. Missing Values

Checking for missing values helps us identify whether any data is unavailable in the dataset.

```
[5]: df.isnull().sum()
```

```
[5]: CustomerID      0
     Age            0
     Tenure_Months  0
     MonthlyCharges  0
     Contract       0
     InternetService 0
     TechSupport    0
     PaymentMethod  0
     Churn          0
     dtype: int64
```

1.5 4. Descriptive Statistics

The `describe()` function provides a statistical summary of the numerical columns, including count, mean, standard deviation, minimum, quartiles, and maximum values.

```
[6]: df.describe()
```

```
[6]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges
count	20.00000	20.000000	20.000000	20.000000
mean	1010.50000	37.700000	24.950000	75.650000
std	5.91608	10.387746	17.479236	16.865646
min	1001.00000	22.000000	2.000000	45.000000
25%	1005.75000	29.750000	11.000000	63.750000
50%	1010.50000	37.000000	22.000000	76.500000
75%	1015.25000	45.500000	37.000000	88.500000
max	1020.00000	56.000000	60.000000	105.000000

1.6 Conclusion

Basic Exploratory Data Analysis was performed on the Customer Churn dataset. The dataset contains 20 rows and 9 columns. The dataset includes both numerical and categorical variables. No missing values were found. The `describe()` function was used to obtain the descriptive statistics of the numerical columns. This basic exploration helps us understand the structure and quality of the dataset before performing further analysis and visualization.